



Cafezal Machine Learning Forecasting Project

Final Technical Report

1. Project Objective

The objective of this project was to **build an in-house machine learning-based forecasting system** capable of predicting future sales at the branch level (daily and hourly), identifying peak periods, and providing actionable insights to optimize staffing, inventory, and operations.

The long-term vision is to establish a **predictive analytics foundation** that Cafezal can grow and integrate into broader strategic decisions.

2. Dataset Overview

- **Sales Data:**
 - Historical POS data (2023 – early 2025)
 - Daily and hourly timestamped transactions
 - Branch-level separation
- **External Weather Data:**
 - 1 year of historical weather in Milan (temperature, humidity, rainfall, windspeed, general conditions)
- **Derived Features:**
 - Rolling averages (7-day rolling mean and standard deviation)
 - Lagged sales values (previous day sales)

- Day-of-week, month, hour-of-day encodings
 - Weather conditions encoding
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3. Methodology

Data Preprocessing:

- Merged and cleaned datasets
 - Extracted key time components (hour, day, month, weekday)
 - Generated lag and rolling features to capture temporal dependencies
 - Merged weather data and filled missing values using forward fill techniques
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Model Development:

- **Model Type:** Random Forest Regressor (for sales prediction)
- **Peak Classification:** Random Forest Classifier (for high-traffic day/hour prediction)
- **Training Approach:**
 - 75/25 Train/Test split
 - KFold Cross-Validation (5 splits) for stability checks
 - Hyperparameter tuning to reduce overfitting (controlled depth, min samples, class balancing)
- **Target Variables:**
 - Daily total sales (for daily model)
 - Hourly total sales (for hourly model)

4. Results Summary

Daily Sales Forecasting (Main Branches):

Branch	Train R ²	Test R ²	Train RMSE (€)	Test RMSE (€)
CAFEZAL_PREMUDA	0.978	0.862	110.48	284.82
CAFEZAL_SOLFERINO	0.973	0.794	79.43	211.46
CAFEZAL_S_GREGORIO	0.947	0.589	43.77	164.11
CAFEZAL_CSO_MAGENTA	0.937	0.481	38.62	121.49

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Observation:

Performance correlates with branch maturity and transaction volume. Larger and older branches like Premuda and Solferino show stronger results.

Hourly Peak Prediction (Premuda Example):

- **Final Daily Peak Classifier F1 Score:** ~78% (Cross-Validation Average)
- **Final Hourly Peak Classifier F1 Score:** ~36% (Cross-Validation Average)
- **Note:**
Hourly peak prediction remains more challenging due to higher noise and variability. Performance is expected to improve as more granular operational and marketing data becomes available.

5. April 2025 Forecasting

- ✓ Generated complete **April 2025 daily sales forecasts** for all major branches
- ✓ Predicted daily sales trendlines based on historical sales, weather patterns, and seasonal effects

Deliverables:

- CSV files containing daily sales forecasts for each branch
 - Visualization plots for easy business interpretation
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6. Technical Improvements Made During Project

- **Overfitting Management:**
Improved classifier generalization by:
 - Controlling model complexity (max depth, min samples)
 - Stratified splitting to preserve class balance
 - Cross-validated evaluation
 - **Feature Engineering Enhancements:**
Integrated external weather data
Added rolling and lagged features to improve temporal sensitivity
 - **Visualization and Interpretation:**
Developed side-by-side plots showing actual vs predicted sales with clear Train/Test splits for transparency
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7. Key Learnings and Next Steps

- **Data Collection Expansion:**
 - Incorporating event schedules, promotions, and marketing campaign data will drastically improve forecast accuracy
 - Real-time app usage data (coming soon) will allow live updating of forecasts
- **Advanced Modeling:**

- Transition to more advanced models (e.g., Gradient Boosting, LSTM networks) once more live data streams are available
 - **Operational Integration:**
 - Embedding forecast outputs into staffing, supply chain, and marketing decision-making processes
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8. Final Remarks

This project demonstrated that **even a basic machine learning model**, using limited external factors, can already provide valuable predictive insights at branch-level resolution.

With continued investment in data collection, modeling, and automation, Cafezal can evolve into a **fully data-driven organization**, maximizing revenue, operational efficiency, and customer satisfaction.